



AI-Powered SupTech for Financial Inclusion

How Emerging Market Authorities Can Unlock Its Potential

June 2026 | Denise Dias and Juan Carlos Izaguirre

As AI rapidly reshapes financial services, financial sector authorities (FSAs) in emerging markets and developing economies (EMDEs) face mounting pressure to oversee AI adoption while leveraging it in their own operations — yet only 53% of EMDEs use AI-powered SupTech compared to 85% in advanced economies. This gap creates both urgency and opportunity. AI-powered SupTech can strengthen three core FSA functions: support, regulation, and supervision, advancing financial inclusion while raising important questions about governance, automation bias, and operational risk. For EMDE FSAs, understanding and acting on this technology shift will require strengthening legal foundations, pursuing realistic digital transformation, enhancing risk management, building adaptive organizational culture, and deepening domestic and international collaboration.

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Executive Summary

As artificial intelligence (AI) rapidly transforms financial services, financial sector authorities (FSAs) in emerging markets and developing economies (EMDEs) are under growing pressure to keep pace by overseeing financial service providers' (FSPs') adoption of AI and leveraging AI in their own work. Yet a significant adoption gap persists. While 85% of FSAs in advanced economies use AI-powered supervisory technology (SupTech), only 53% of FSAs in EMDEs do so.

EMDE FSAs can leverage three mutually reinforcing advantages of AI-powered SupTech: operational efficiency, broader analytical reach, and augmented decision-making. At the same time, adoption introduces key risks that FSAs must actively manage: AI governance and accountability risks, automation bias, and operational and technology risks.

This brief argues that AI-powered SupTech offers FSAs in EMDEs a strategic opportunity to transform their capabilities across three functions:

- support (e.g., data quality improvements, process automation);
- regulation (e.g., impact assessments, process streamlining); and
- supervision (e.g., earlier risk detection and market monitoring).

Each of these can advance financial inclusion directly by allowing FSAs to monitor and disseminate richer information, and indirectly by reducing compliance costs for FSPs, thereby increasing trust in the financial system, and supporting FSA mandates that, in turn, contribute to inclusive finance.

This brief proposes five priority recommendations for EMDE FSAs to fully leverage AI-powered SupTech in a manner that fosters a responsible and inclusive financial ecosystem:

1. strengthen legal foundations for data protection and ethical AI use;
2. pursue an ambitious but realistic digital transformation agenda;
3. enhance AI risk management;
4. transform organizational culture; and
5. leverage domestic and international collaboration.

Introduction

Artificial intelligence (AI) is rapidly transforming financial services—a development to which financial sector authorities (FSAs) in emerging markets and developing economies (EMDEs) must both respond and, where possible, adapt. Financial service providers (FSPs) are using AI in both customer-facing and back-office functions (IIF and EY 2025, McKinsey 2025). A large global survey indicates that 65% are actively using AI and 25% are assessing AI solutions or pilot projects (Nvidia 2026), while global AI-related spending in the financial sector is projected to grow from \$35 billion in 2023 to \$97 billion by 2027 (WEF 2025). The scale and speed of this transformation create both an imperative and an opportunity for FSAs to adopt AI to enhance their supervisory capacity and better oversee FSPs' use of AI.

AI holds significant promise for advancing responsible financial inclusion and addressing persistent barriers faced by low-income individuals and micro and small enterprises (MSEs). This potential stems from AI's capacity to rapidly collect, process, and learn from vast volumes of diverse data. In Brazil, for instance, FSPs are using AI to expand MSE credit by analyzing transaction data from the fast payment system Pix, through open finance—an innovation that also illustrates the new supervisory challenges FSAs face (Dias 2025). However, AI can also amplify or introduce risks, including demographic biases embedded in data (Sankar 2025), the exclusion of segments with limited digital access, such as women and young girls (Poggi 2025), fraud and data misuse (Chalwe-Mulenga and Duflos 2026; Duflos 2025), consumer agency and liability concerns (Quick et al. 2025), and prudential risks, including risk concentration, herding behavior, and entrenched market dominance (T. Zhang 2025; Sankar 2025; Leitner et al. 2024; FSB 2024).

FSAs in EMDEs face significant capacity pressures from multidimensional transformations—rapid digitization, evolving business models, geopolitical instability, and climate change—that stretch their resources and attention (Zetzsche et al. 2020). Against this backdrop, FSAs may find it particularly difficult to strike a balance between the opportunities and risks associated with FSPs' adoption of AI, while traditional regulatory and supervisory approaches designed for static, siloed, bank-centric systems (B. Z. Zhang 2025) are ill-suited to this challenge. This creates an increasingly urgent need for FSAs to develop their own AI-powered supervisory capacity to assess FSPs' responsible adoption of AI and harness AI as a tool to enhance supervision and risk identification. When FSAs can conduct better supervision, they are better positioned to ensure that AI-powered financial services reach underserved customers responsibly. Simultaneously, FSAs' effective AI-enabled transformation must encompass fundamental changes in processes, culture, and mindset—not just IT.

Building on the literature on AI in finance, this brief highlight the opportunities, challenges, and risks associated with the adoption of AI-powered supervisory technology (SupTech)^[1] by FSAs in EMDEs to advance financial inclusion. After brief descriptions of the state of adoption and use cases for AI-powered SupTech, the report addresses three questions: (i) What opportunities does AI-powered SupTech offer to support financial inclusion? (ii) What implementation constraints and risks do FSAs in EMDEs face when adopting AI-powered SupTech? (iii) What can FSAs in EMDEs do to leverage AI-powered SupTech? This report aims to assist FSAs in EMDEs on their journey of AI adoption to strengthen their capabilities—a key component of responsible digital finance ecosystems (Duflos et al. 2024).

¹ We refer to SupTech as technology used by FSAs to improve effectiveness and efficiency across their functions. The term was coined by Menon (2017) and adopted by others (e.g., CCAF n.d.; di Castri et al. 2019; World Bank 2018; Dias 2018).

State of Adoption of AI-Powered SupTech

The 2008 global financial crisis exposed significant weaknesses in supervisory data and analytical capacity, prompting a surge in reporting requirements imposed on FSPs. As reported data volumes grew, so did the analytical burdens on FSAs, leading the Basel Committee on Banking Supervision (BCBS) and the Financial Stability Board (FSB) to recommend that supervisors explore new technologies to keep pace (BCBS 2016; FSB 2020).

SupTech adoption has remained uneven since then. According to the Cambridge SupTech Lab and Digital Transformation Solutions (CSL and DTS 2025), 197 FSAs across 140 countries now have at least one live SupTech tool, yet a notable gap persists: 85% of FSAs in advanced economies use SupTech, compared with 53% in EMDEs. A World Bank (2025) survey of 27 FSAs in EMDEs, including 17 in Africa, notes that AI adoption is a board-level priority for most authorities.

FSAs' interest in AI-powered SupTech is accelerating, especially in EMDEs. CSL and DTS (2025) shows that 30% of the 120 surveyed EMDE FSAs are piloting or deploying AI-powered tools on a limited basis, compared with 17% in 2024, while use of generative AI (GenAI) more than doubled between 2023 (8%) and 2025 (18%). EMDE FSAs are gradually moving from early AI exploration toward plans to use AI agents and GenAI for tasks such as internal knowledge management, complaint analysis, and risk and compliance assessments (World Bank 2025).

Notably, none of the African FSAs surveyed by the World Bank have yet implemented AI-powered SupTech for core supervisory functions. The reported benefits are already tangible: most EMDE FSAs surveyed by CSL and DTS (2025) indicate that SupTech has effectively automated and streamlined supervisory work (71%), improved data accuracy and reliability (64%), and enhanced risk detection and mitigation capacities (62%). Expanding these gains will depend on FSAs pursuing a balanced mix of AI use cases, accounting for risks as carefully as they assess opportunities. This expansion is imperative in a context where 32% of FSPs and only 10% of FSAs report advanced AI adoption in EMDEs, while 65% of all FSPs and 59% of all FSAs do not monitor AI for discrimination, exclusion, or systemic bias (CCAF 2026). AI can enable FSAs to assess and act on risks and outcomes across consumer segments, thereby strengthening risk-based supervision and financial inclusion efforts.

Use Cases for AI-Powered SupTech

AI-powered SupTech can be deployed across three types of FSA functions—support, regulation, and supervision—each with specific use cases (see Figure 1 and Annex 1):

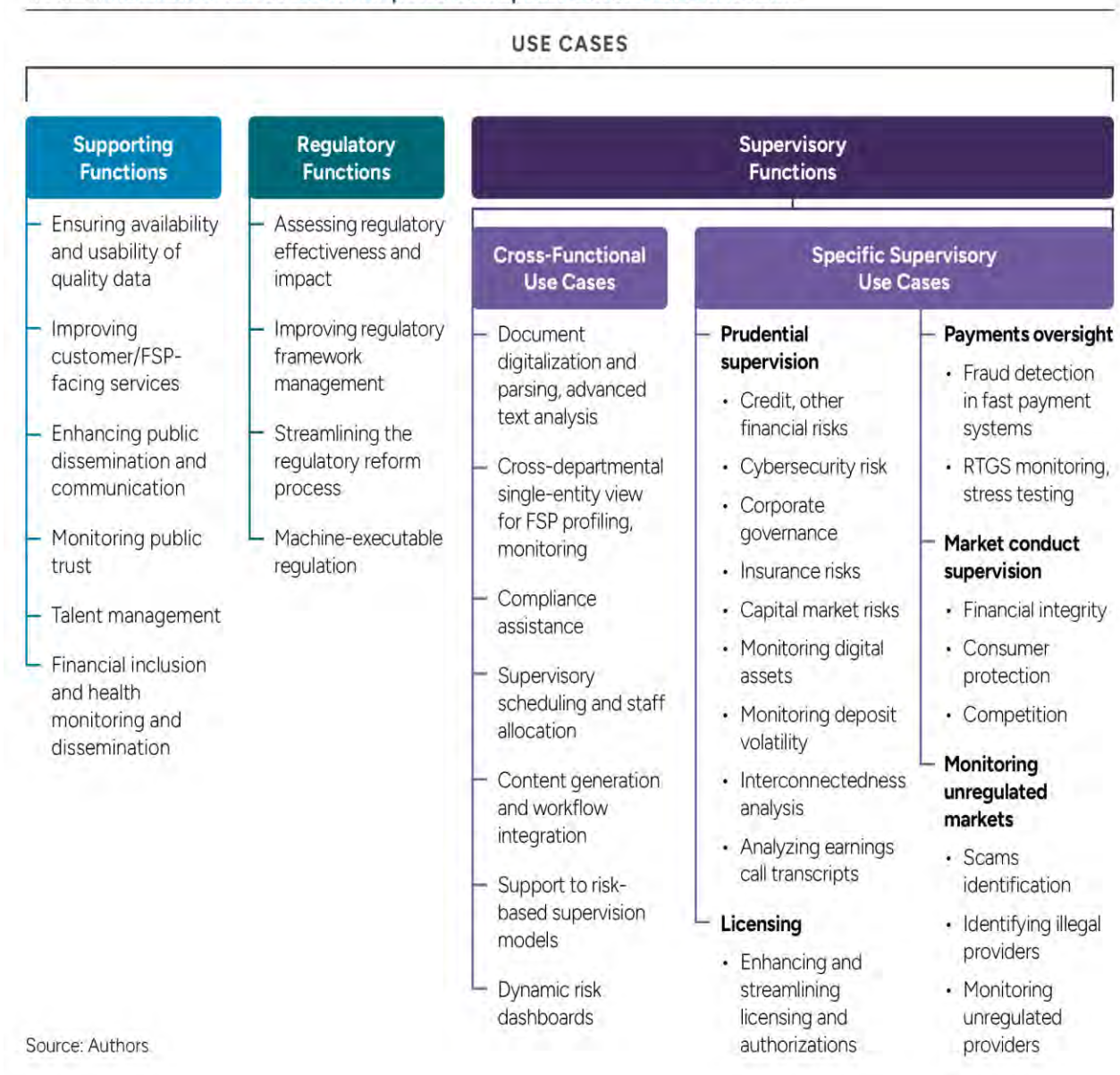
1. **Support Functions:** Beyond core supervision and regulation, SupTech can enhance a range of FSA support functions. These include improving data availability and usability (collecting granular data from FSPs and their customers, standardizing and transforming reported supervisory data, and digitizing or automating operational processes) as well as strengthening cybersecurity and supporting talent management by addressing skill gaps and identifying unconscious biases that may affect supervision. As Dohotaru et al. (2025) note, productivity gains are anticipated across all supervisory processes involving unstructured data or large volumes of data. Support functions also encompass activities directly relevant to financial inclusion, such as public trust monitoring, complaint handling, and financial inclusion and health monitoring. While these areas are underrepresented in the SupTech literature, they carry significant implications for inclusive finance. Stronger data infrastructure and analytical capacity enable FSAs to better detect disparities in access, usage, quality, and risk across consumer segments, making supervision more proportionate, risk-based, and customer-centric, and helping FSAs manage tradeoffs and synergies among inclusion, stability, protection, and integrity (Tomilova and Valenzuela 2018).
2. **Regulatory Functions:** SupTech can enhance assessments of regulatory impact and effectiveness, including by tracking financial health outcomes and identifying how regulatory decisions affect underserved populations (Izaguirre 2020, GPFI 2024). SupTech also supports the development of regulatory frameworks by helping FSAs identify international good practices, incorporate supervisory insights, and detect linkages or overlaps with existing regulations. It can also streamline reform processes by systematically processing and incorporating feedback from diverse inclusive finance stakeholders during consultations or by surfacing key themes from consumer complaints and industry engagement. Finally, FSAs can use tools such as machine-executable regulation, GenAI, and AI agents to improve the clarity

and consistency of regulatory language², reducing ambiguity and lowering compliance costs for FSPs serving low-income customers.

3. **Supervisory Functions:** Most of the literature focuses on this area, but there is no single agreed taxonomy or catalogue of use cases (CCAF n.d.; BIS 2025; Prenio 2025; Dohotaru et al. 2025; Prenio et al. 2024; RegTech Association 2024; de Souza Neves Lopes et al. 2021; FSB 2020). These functions include both cross-cutting processes and core supervisory use cases. The former relate to administrative, procedural, statistical, and analytical support that helps supervisors manage resources, activities, data, and reports more effectively. The latter relate to the direct use of SupTech techniques and tools in supervisory analysis across mandates and sectors. The combination of greater availability of comprehensive data and enhanced efficiency will enable FSAs to assess the financial inclusion implications of their work and continuously identify data quality gaps, reporting them to FSAs' support functions to enhance data.

² Modern AI systems are increasingly capable of accurately reading, interpreting, and reasoning over texts drafted for humans without the need for prior translation into a machine-readable format. As a result, the incremental value of machine-readable regulations is likely to decrease, while AI tools make language clearer and native AI-readable.

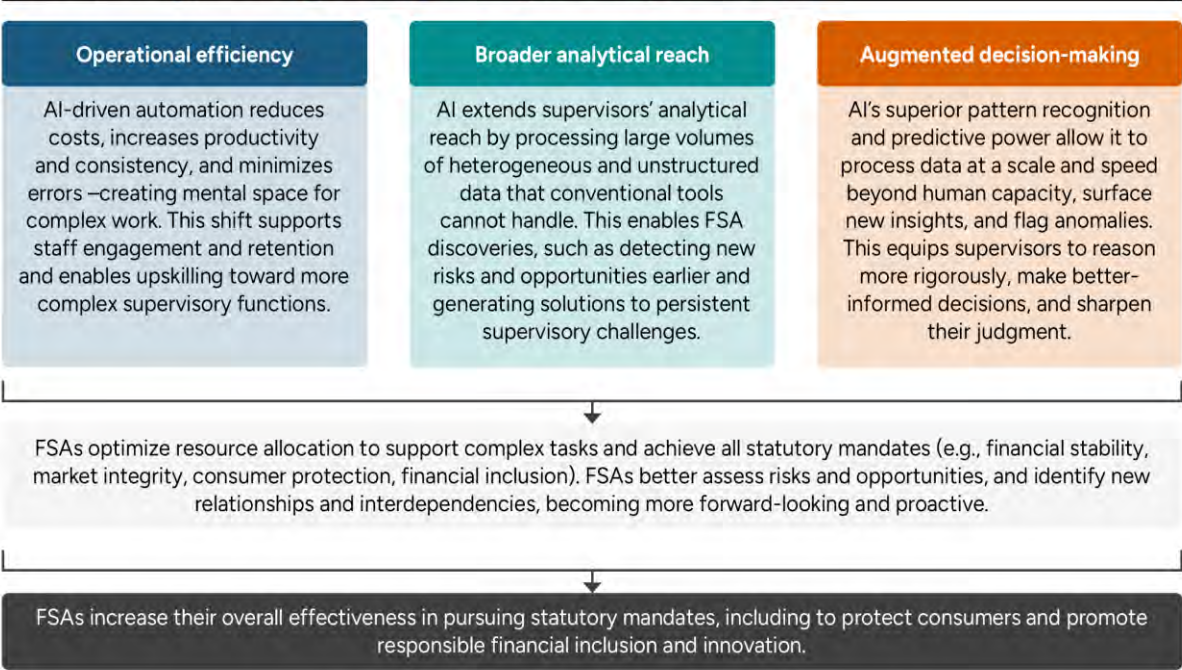
FIGURE 1. Illustrative Use Cases for AI-powered SupTech Across FSA Functions



AI SupTech and Financial Inclusion Opportunities

AI-powered SupTech’s transformative potential rests on its foundational ability to rapidly collect, process, and learn from vast volumes of varied data. Rooted in this core capability, there are three mutually reinforcing advantages that enable FSAs to establish stronger foundations for forward-looking supervision, support their mandates, and ultimately foster responsible financial inclusion and innovation: (i) operational efficiency, which automates routine tasks, reduces costs, and minimizes errors; (ii) broader analytical reach, which enables earlier detection of risks and opportunities by processing large volumes of previously underused data; and (iii) augmented decision-making, which leverages AI’s analytical capabilities to enhance supervisors’ ability to make better-informed decisions (Figure 2).³

FIGURE 2. **Mutually Reinforcing Advantages of AI-powered SupTech for FSAs**



Source: Authors

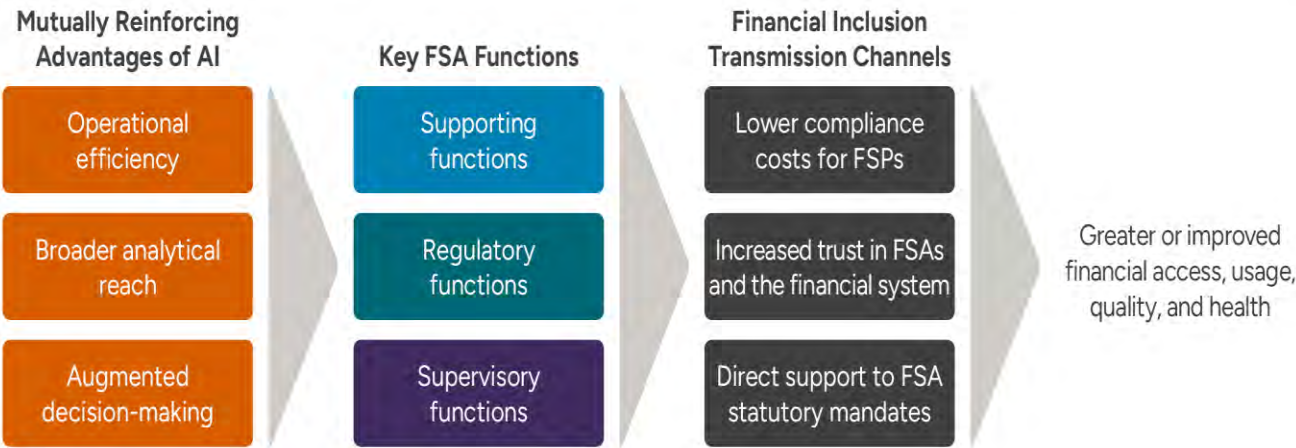
³ FSPs may also leverage AI to better serve and understand customers, discover markets, and reduce the cost of misplaced risk aversion (Caire and Fernandez Vidal 2024; Gruver et al. 2024; Kruijff et al. 2024; Shrivastava and Harris 2025).

The clearest pathway for AI-powered SupTech to support financial inclusion is through direct improvements in the monitoring and dissemination of financial inclusion and financial health, as a part of FSA's support function. Improvements in FSAs' support functions are especially important because they include better availability and usability of quality data, as well as the staff capabilities needed to use such data across all FSA activities. However, AI can also indirectly support financial inclusion through a range of use cases across FSAs' support, regulatory, and supervisory functions.

The financial inclusion linkages of AI-powered SupTech are transmitted through three channels that are not mutually exclusive and can reinforce one another to foster a more responsible and inclusive financial sector:

- i) Lower regulatory compliance costs for FSPs by reducing the costs of submitting regulatory reports and checking FSA databases (e.g., for creditworthiness or customer due diligence).
- ii) Increase trust in FSAs and in the financial system among consumers who are self-excluded from formal finance. According to the World Bank's Global Findex, lack of trust in financial institutions discourages about 20 percent of adults without accounts across low- and middle-income economies (Klapper et al. 2025).
- iii) Provide direct support for FSAs' statutory mandates and responsibilities, which in turn may advance or constrain financial inclusion efforts (see Figure 3 and Annex).

FIGURE 3. **AI-Powered SupTech for Key FSA Functions and Financial Inclusion Linkages**

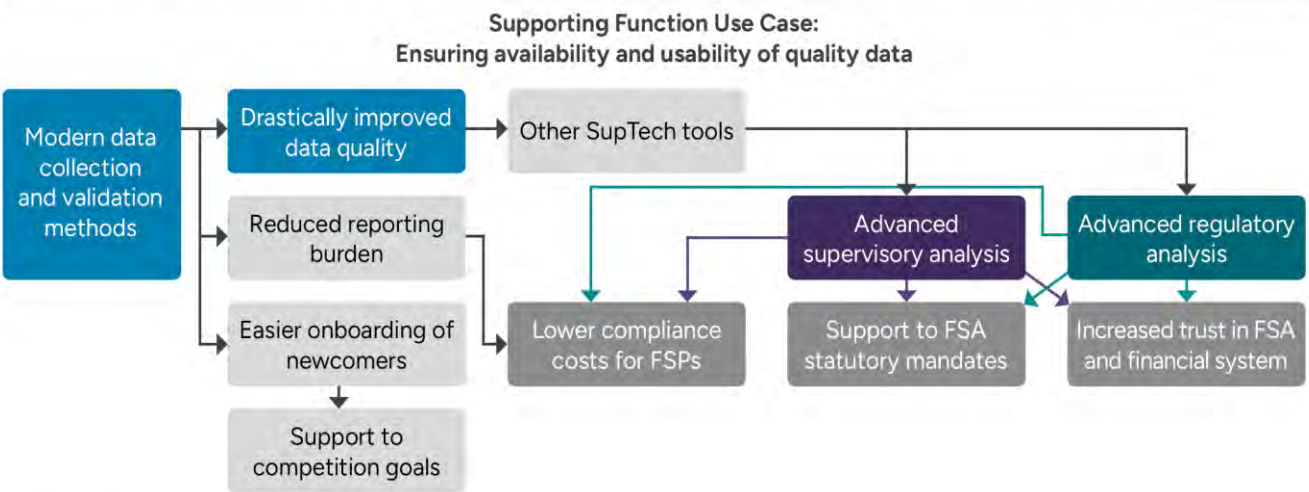


Source: Authors

Linkages between FSA Support Functions and Financial Inclusion

AI-powered SupTech deployed in FSA support functions, such as data collection and management, can enhance other use cases, amplifying its impact on financial inclusion. For example, AI-powered data collection methods can expand the volume of granular and unstructured data available to FSAs, enabling them to better assess customer risks and outcomes, including for vulnerable segments, across all FSA mandates. AI-powered validation methods can also significantly improve the quality of supervisory data. Better and more comprehensive data can enhance other AI-powered SupTech applications, such as advanced analytics that support statutory goals, ultimately contributing to greater public trust in FSAs. Reduced compliance costs could result from optimized reporting and data validation, along with more efficient and precise risk assessments. Better reporting systems could also speed up FSP authorization, supporting competition (see Figure 4). Additionally, some FSA support functions relate to financial inclusion, such as public trust and financial health monitoring and complaints handling.

FIGURE 4. Example of Financial Inclusion Linkages with AI-Powered SupTech in Supporting Functions



Source: Authors

Linkages between FSA Regulatory Functions and Financial Inclusion

AI-powered SupTech applied to regulation can enhance the transparency, clarity, and certainty of FSAs' regulatory frameworks, reducing challenges and costs for both supervisors and FSPs. Beyond the regulatory drafting phase, SupTech supports a more dynamic regulatory cycle that is better informed by supervisory findings, market developments, consumer risks and outcomes, and inter-institutional coordination. This improved cycle enables more evidence-based, data-driven, proportionate, and

proactive regulation, which may strengthen public trust in FSAs and reduce FSP compliance costs, both of which can support financial inclusion. Additionally, AI could help FSAs review regulatory frameworks to identify and address provisions that may be biased or have exclusionary or discriminatory effects.

Linkages between FSA Supervisory Functions and Financial Inclusion

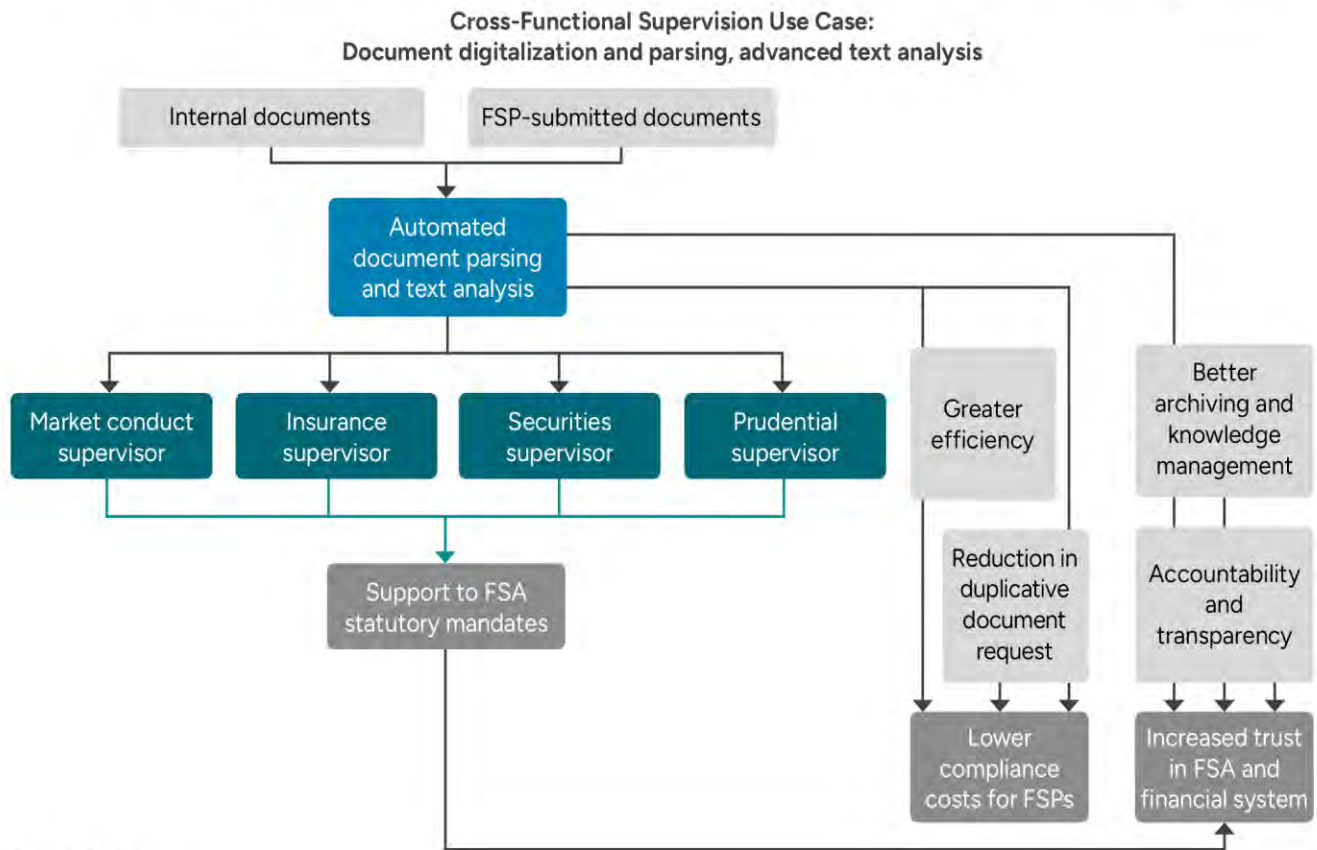
The linkage between AI-powered SupTech adoption and financial inclusion through the fulfillment of FSA mandates and goals has been a focus of the SupTech literature. For instance, FSB (2025, 2020) highlights that SupTech can help FSAs strengthen the stability and resilience of the financial system by improving oversight and supervision. This includes improvements in identifying and assessing risks, allocating supervisory resources, and adopting risk mitigation solutions that contribute to broader access to and usage of quality financial services.

The link is much clearer when SupTech is used for competition and consumer protection. For instance, a SupTech application that increases the efficiency of licensing, such as Bank of Ghana's Online Regulatory and Analytical Surveillance Software (ORASS), could contribute to competition, which supports financial inclusion (Kumaraswamy and Kremnitzer 2025; Dias 2025; Teimory et al. 2025). Another example is AI-powered social media monitoring that assesses consumer risks and detects early warning signals. For example, the CGAP/Reserve Bank Innovation Hub pilot in India's digital credit market used natural language processing to identify unregulated apps that were harmful and subsequently added to a watch list (Duflos et al. 2023). Similarly, authorities like the Australian Securities and Investments Commission use AI-powered SupTech tools to spot illegal and misleading online promotions and unlicensed finfluencers, triggering takedowns and alerts for the benefit of consumers (Chang et al. 2026).

AI-powered SupTech tools deployed across supervisory departments can also amplify effects on financial inclusion. For example, an AI application for document parsing and text analysis could help supervisors develop other AI-enabled tools to automate repetitive analyses of internal documents and FSP submissions (e.g., consumer contracts, fee schedules, board meeting minutes, prospectuses of investment funds, inspection reports), supporting various supervisory mandates. These tools enable FSAs to identify discriminatory provisions in FSP documents and consistent unfair FSP treatment of customer segments. A greater ability to meet supervisory objectives would enhance public trust in FSAs. Cross-departmental sharing of document analysis tools would lead to greater efficiency and fewer duplicative requests, thus reducing compliance costs for FSPs. Document parsing tools could also improve archiving and knowledge management and enhance accountability and transparency, further increasing trust in FSAs (Figure 5).

Furthermore, AI-powered SupTech can strengthen efforts in data standardization and quality that underpin data-sharing regimes while enabling better supervision of open finance (Dias et al. 2026).

FIGURE 5. **Example of Financial Inclusion Linkages with AI-Powered SupTech in Cross-Functional Supervision**



Source: Authors

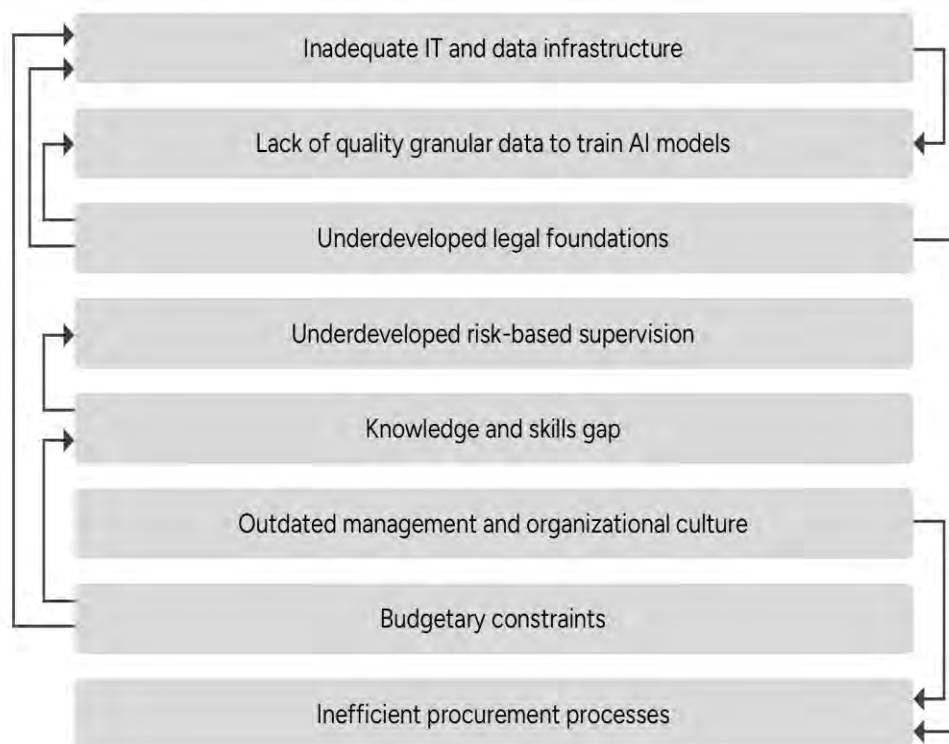
Enabling the Adoption of AI-Powered SupTech in EMDEs

FSAs in EMDEs face many constraints in leveraging and implementing AI-powered SupTech due to mutually reinforcing foundational weaknesses (Figure 6). Addressing these weaknesses will be essential to effective and sustainable adoption of AI-powered SupTech:

Inadequate IT and data infrastructure: Integrating AI into existing IT infrastructure is the main challenge to AI adoption among FSAs in EMDEs, many of which are encumbered by legacy IT systems and inefficiencies in data collection and management (World Bank 2025). They often lack the computational power required to deploy AI models, and their limited adoption of cloud computing further constrains AI adoption. With respect to regulatory reporting, even FSAs in advanced economies face challenges, such as fragmented reporting frameworks that hinder reusability and result in duplicative and inconsistent data requests to FSPs (Bank of England 2020). Inadequate infrastructure to handle high volumes of granular data limits EMDE FSAs' capacity to conduct market monitoring in areas such as fraud, digital credit, and women's financial inclusion (Izaguirre et al. 2022; Alonso and Dezso 2023).

Lack of quality granular data to train AI models: Most FSAs in EMDEs lack the high-quality granular data needed to train AI models. Additionally, they often lack the capacity to combine structured and unstructured data from different internal and external sources so they can fully leverage AI capabilities. Crucially, data also needs to be inclusive: it should represent FSPs serving underserved customer segments and contain attributes that enable segmented analysis by sociodemographic characteristics such as gender, age, and income (Alonso and Dezso 2023; Salman et al. forthcoming).

FIGURE 6. **Barriers to Adoption of AI-Powered SupTech in EMDEs**



Source: Authors

Underdeveloped legal foundations: AI adoption by FSAs and FSPs requires robust legal frameworks. More than 60% of FSAs in EMDEs cite data protection, privacy, and security concerns as top barriers to AI adoption (CSL and DTS 2025). The absence of a data protection framework limits the use of granular data and AI-powered SupTech. In addition, inadequate laws governing FSAs’ procurement processes reduce their ability to use cloud computing; 35% of EMDE FSAs surveyed by the World Bank (2025) face legal barriers to using cloud services. AI-powered SupTech may also disproportionately benefit incumbents if outdated frameworks leave inclusive market players outside the regulatory perimeter.

Underdeveloped risk-based supervision: A significant barrier to FSAs’ effective AI adoption, often overlooked in the literature, is the immature state of risk-based supervision (RBS) in many EMDEs (Adrian et al. 2023), which undermines the fulfillment of their mandates. Most FSAs in EMDEs have adopted an RBS approach, but only a few have fully operationalized it by transforming their supervisory practices, skills, and expertise. This limitation affects FSAs’ ability to leverage AI-powered SupTech fully and manage associated risks (see next section). At the same time, Dohotaru et al. (2025) note that AI

can enable RBS by helping FSAs make necessary transitions, such as automating repetitive tasks that do not require supervisory judgment.

Knowledge and skills gap: FSAs in EMDEs face significant challenges attracting and retaining talent with the skills to leverage AI-powered SupTech. BIS (2024) highlights that nearly 90% of the central banks surveyed found that recruiting staff has become more difficult over the last five years, especially in IT, cybersecurity, fintech, data science, and AI/machine learning. Among EMDEs, CSL and DTS (2025) shows that limited internal IT capacity and data analytics skills were cited by 41% and 40% of FSAs, respectively, as barriers to implementing SupTech. The problem is compounded by the need to combine strong supervision expertise with new technical skills.

Outdated management and organizational culture: Many FSAs' management and organizational cultures are not conducive to innovation and technology adoption due to the lack of top management support, data silos, limited inter-departmental collaboration, general resistance to change, and conservatism. For example, issues of financial inclusion often fall outside the scope of supervisors' work. Hence, they often fail to appreciate the relationship between their core work and financial inclusion (e.g., how FSPs' over-compliance with anti-money laundering/countering the financing of terrorism (AML/CFT) rules can lead to exclusion and increase risks at the national level). CSL and DTS (2025) shows that non-supportive institutional cultures are a bigger challenge for FSAs in EMDEs (40%) than in advanced economies (32%).

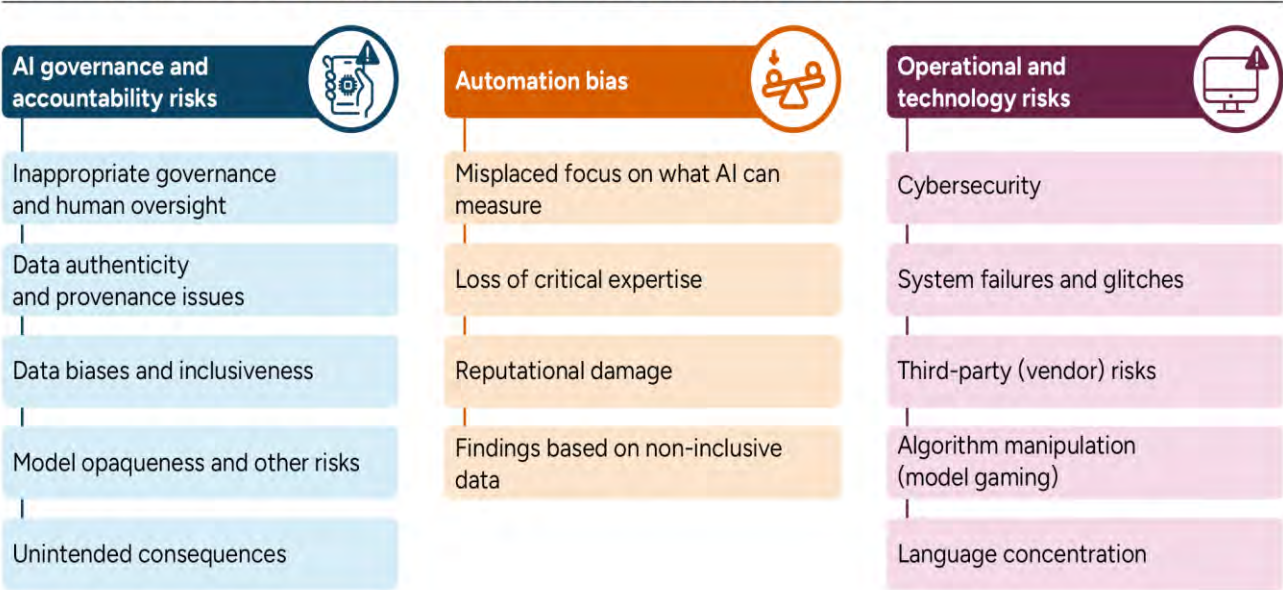
Budgetary constraints: Budgetary constraints impact all aspects of SupTech adoption and maintenance, from talent acquisition to IT and data infrastructure reforms. Over one-third (36 percent) of FSAs report a lack of dedicated SupTech budgets, while 60 percent of EMDEs express concerns about inadequate budgets, compared to 36 percent in advanced economies (CSL and DTS 2025). Considering the weaknesses in the legacy systems and data architectures of many FSAs in EMDEs, the upfront costs of establishing strong foundations for AI-powered SupTech integration can be significant and require careful planning.

Inefficient procurement processes: CSL and DTS (2025) found that procurement processes are a significant barrier to the development and deployment of SupTech, identified by 39 percent of all FSAs and 34 percent in EMDEs as a major problem, even when financial resources are available.

Preventing Risks in Adopting AI-Powered SupTech

The adoption of AI-powered SupTech by FSAs in EMDEs also poses risks, underscoring the need for a cautious approach. Addressing risks preemptively supports the long-term, sustainable, and effective implementation of AI-powered SupTech and helps prevent negative outcomes. Figure 7 summarizes these key risks, most of which are not unique to EMDEs or to a financial inclusion context.

FIGURE 7. Key Risks Faced by EMDE FSAs when Adopting AI-Powered SupTech



Source: Authors

AI governance and accountability risks: Large but potentially incomplete, outdated, or biased data, as well as challenges with models and AI usage, can create risks for FSAs, especially in EMDEs—most of which are aware of these risks but have not addressed them yet (World Bank 2025).

- Inappropriate governance and inadequate human oversight. If AI-driven FSA decisions are seen as opaque, inconsistent, or biased, public confidence in the fairness and competence of FSAs could be eroded, which could have financial stability and inclusion consequences. Decisions derived from AI analysis without adequate human oversight may expose FSAs to risks, including legal challenges.
- Complexity in ensuring data authenticity and provenance. The confidentiality and provenance of training data are critical considerations when AI-powered SupTech models are trained with FSP data (FSB 2020). Questions may arise about the legitimacy of the data used and the extent to which FSPs have consented to or understood the use of their data.
- Risks associated with biases and data inclusiveness shortcomings. AI outputs are jeopardized when input datasets embed patterns of discrimination and structural biases—such as historic disparities across genders, income groups and races in deposit-taking, lending, and insurance provision. Input data may also exclude relevant segments of the financial sector, such as FSPs catering to underserved customers, which may not be subject to reporting requirements, or may report comparatively lower quality data. Moreover, large language models (LLMs), which many FSAs in EMDEs are using (World Bank 2025), are trained on generalized data sources that may not be suitable for specialized FSA purposes. These issues limit the usefulness of AI outputs and can perpetuate exclusion by making FSAs blind to disparities in access, usage, and quality of financial services, and unable to identify and act upon linkages between inclusion, financial health, and other mandates.
- Complexity in model risk management can lead to opaqueness, model malfunction, errors, and contagion. Model risk management is more complex with AI, especially when using "black box" models that create a trade-off between predictive accuracy and explainability. Lack of explainability is a serious issue for any FSA because they must justify their actions and maintain transparency and accountability. Performing actions based on opaque AI systems can result in legal challenges and damage to reputation. Further, AI models can fail, malfunction, have design errors, or deteriorate over time, and LLMs may have "hallucinations." The expected growth in the use of automated systems using AI Agents will generate additional risks, such as cascade failures across AI systems due to algorithmic contagion or collusion, i.e., when systems learn from and amplify each other (T. Zhang 2025).
- Unintended consequences. AI-powered SupTech may not only have positive effects but also inadvertent negative consequences for financial inclusion. For example, Degryse et al. (2025) found that the Central Bank of Brazil's use of SupTech for prudential supervision revealed inconsistencies in risk reporting. This finding led banks to tighten credit to less creditworthy firms, negatively affecting the economic performance of such borrowers, even though the supervisor did not require changes to risk-taking.

Automation bias: Automation bias, or data blindness, occurs when humans over-rely on AI tools and defer to AI outputs or recommendations without critically questioning them or the strength of the

underlying training data. Automation bias can affect supervisory judgment and lead to the loss of institutional knowledge and expertise (World Bank 2025; Dohotaru et al. 2025). Overreliance on AI is particularly problematic in contexts where input data does not cover all the areas that a human-driven approach would, potentially leading to misplaced supervisory focus. This is an important risk for data-scarce markets, where training data may not cover relevant segments or may cover them only imperfectly because of embedded biases. The risk may also be greater in EMDEs, where supervisory capacity and skills are often more limited.

Operational and technological risks:

- Cybersecurity risk is amplified by the sheer volume of data and because AI systems connected to external sources present new attack vectors. For instance, there could be prompt leakages, where sensitive prompts using GenAI for supervisory analysis are revealed.
- System failures and glitches in automated processes could disrupt time-sensitive supervisory activities, leading to delays or flawed risk assessments.
- Third-party risk is created when vendors are involved. Vendor lock-in risk can increase dependency, reduce flexibility and, depending on the vendor's location, undermine data sovereignty or conflict with data localization rules, which are more common in EMDEs. Vendors may also increase data privacy risks. Vendor risks are exacerbated by the concentration of cloud and AI services in a few companies, especially in EMDEs where most FSAs rely on outsourcing to a small set of global vendors (World Bank 2025). When the same vendors serve both FSAs and FSPs, cyberattacks, outages, data breaches, and other risks could simultaneously affect the supervisor and the supervised entities, creating systemic vulnerabilities.
- Algorithm manipulation (model gaming) is the risk of external actors gaming SupTech systems by "poisoning" training data (BIS 2024). Fraud detection, compliance monitoring, risk scoring, and external-facing systems are particularly vulnerable. For example, Dohotaru et al. (2025) note the risk of chatbot misuse to find regulatory loopholes and prompt injection attacks via external-facing GenAI, which may generate content with legal consequences for FSAs.
- Language concentration occurs because most leading AI models are predominantly trained on English-language datasets. This concentration may increase risks of misinterpretation, inaccuracy, and bias, especially when the models are applied in different contexts and local languages are underrepresented in training datasets.

Recommendations

Below is a list of high-priority recommendations for FSAs in EMDEs seeking to adopt AI-powered SupTech in support of financial inclusion (Figure 8). They align with proposals by the World Bank (2025), Dohotaru et al. (2025), and Bains et al. (2025) regarding overall AI adoption by FSAs. Each FSA will need to assess its own context to adapt and apply these recommendations effectively. The recommendations focus on creating the underlying conditions for AI adoption, which are more often lacking among FSAs in EMDEs than among those in advanced economies. They also emphasize the need for FSAs to mitigate AI adoption risks effectively as a central area for improvement.

FIGURE 8. Recommendations to Harness AI-Powered SupTech in Support of Financial Inclusion

1 Legal Foundations	2 Digital Transformation	3 AI Risk Management	4 Organizational Culture	Domestic and International Collaboration
• Data protection and privacy • AI ethics • Permitted use of granular data • Permitted use of cloud computing • Adequate regulatory perimeters covering inclusive FSPs • Adequate reporting requirements for inclusive FSPs	• Data strategy implementation for modern IT and data infrastructure to enable high-quality granular data • SupTech strategy implementation to transform supervisory processes, aligned with workforce strategy • Data and AI governance frameworks appropriate for the use of granular data and AI tools	• AI (or model) risk management framework • Prioritization of explainable and interpretable models • Model performance evaluations • Specialized risk mitigation tools • Inclusive and comprehensive training data	• Workforce strategy leveraging multidisciplinary teams and supported by SupTech strategy • Innovation and responsible financial inclusion culture • Foundational supervisory approach explores AI-powered SupTech as catalyst for risk-based supervision	• Domestic, regional, and global platforms to find solutions to challenges of AI-powered SupTech, risks of vendor concentration • Shared resources

Source: Authors

1. STRENGTHEN LEGAL FOUNDATIONS

FSAs' deployment of AI must be supported by legal frameworks for data privacy and protection, confidentiality and secrecy, and the ethical use of AI. Such frameworks should align with principles of transparency, fairness, and accountability while protecting personal data. Global AI governance principles include the UN principles for the ethical use of AI (UNSCEB 2022) and the OECD AI principles (OECD 2024). At the same time, these frameworks should not unduly restrict FSAs' use of granular data, including customer-level disaggregated data. The legal framework should also permit FSAs to use cloud services, which may form part of their data strategy (see next recommendation).

To ensure that adoption of AI-powered SupTech is based on inclusive data and benefits all customer segments, some FSAs may need to review their regulatory perimeter to include relevant market players and subject them to appropriate reporting obligations. Prioritizing these foundations will not only reduce long-term risks but also enable cost-effective, staged AI deployment in resource-constrained environments. Dohotaru et al. (2025) note that FSAs can explore AI opportunities while overcoming resource limitations, and Bains et al. (2025) note that experience with small-scope projects helps FSAs establish a learning curve for large-scale implementation.

2. PURSUE AN AMBITIOUS BUT REALISTIC DIGITAL TRANSFORMATION AGENDA

Digitization is both a prerequisite for and an enabler of deriving value from AI-powered SupTech. Some FSAs may need comprehensive digital transformation agendas with three interlinked components: implementation of a data strategy, implementation of a SupTech strategy, and creation of a data and AI governance framework. One example is the Central Bank of the United Arab Emirates' digital transformation strategy, comprising such programs as the Future-Ready Central Bank or Future Hub initiative -- aimed at embedding long-term preparedness and forward-looking decision-making across the organization to make better strategic decisions and visioning -- and the Financial Infrastructure Transformation Programme -- aimed at accelerating the digital transformation of the financial sector, including SupTech, financial cloud, and innovation hub initiatives (CBUAE 2025).

- **Data strategy implementation:** Quality granular data are crucial for AI-powered SupTech deployment. It is therefore critical for FSAs to modernize regulatory reporting systems and the related IT infrastructure that enable them to handle granular data. A modern data strategy addresses issues such as reporting automation, including real-time reporting, data interoperability and standardization, secure data sharing, scalability of reporting systems, cloud services, cybersecurity, large-scale data ingestion and integration tools, and data storage systems. For some FSAs in EMDEs, major data reforms may offer an opportunity to leapfrog incremental upgrades, by fully transforming their IT and data infrastructure and, consequently, their regulatory and supervisory processes.

- **SupTech strategy implementation:** A SupTech strategy should be a core pillar of a digital transformation agenda because it facilitates SupTech deployment (CSL and DTS 2025; Gambacorta et al. 2025). Such a strategy helps FSAs plan deployments based on pain points in regulatory and supervisory processes and broader objectives (Izaguirre et al. 2022). The SupTech strategy should align with and support the workforce strategy, as the AI models adopted by an FSA will shape workforce needs (Bell et al. 2025). For example, in some areas, FSAs may adopt low-code SupTech tools, which do not require users to know coding languages, or use AI to translate plain-language queries into code. This can reduce reliance on scarce talent, ease staffing constraints, and accelerate AI adoption.
- **Data and AI governance framework:** A robust data and AI governance framework includes principles for data collection, management, quality, consistency, transparency, fairness, privacy, and ethical use, alongside accountability mechanisms anchored in the human-in-the-loop principle (human oversight of AI models). It should include protocols for data provenance and lineage, such as metadata recording and master data catalogs, as well as documentation requirements covering logs, version histories, and decision rationales for AI tools and their outputs. It should also include ethics assessments for high-risk deployments and principles governing vendor use that balance cost considerations with accountability obligations (FSB 2020). FSAs should also consider appointing senior leaders with dedicated responsibility for institutional AI governance.

3. ENHANCE AI RISK MANAGEMENT TO MITIGATE BIAS AND OTHER RISKS

FSAs should establish an AI, or model, risk management framework that prioritizes the use of explainable and interpretable models and helps manage AI projects (see Bains et al. 2025). FSAs should adopt a human-in-the-loop model and may use risk-mitigation tools that increase model reliability and address exclusion and unfairness, such as de-biasing techniques and analysis of model outputs disaggregated by customer groups, including by gender. FSAs can also conduct periodic model performance evaluations, algorithmic audits, modeling of failure scenarios (e.g., hallucinations, model drift, miscommunication among AI agents, contagion), and stress tests to ensure model robustness and identify dependencies across AI systems (Barefoot and Kibria 2024). Randomized audits, rotating model features, and adversarial training can all help mitigate model-gaming risk. Other tools are also available: some FSAs restrict AI to specific use cases, while others develop contingency arrangements (World Bank 2025).

4. TRANSFORM ORGANIZATIONAL CULTURE

The effective adoption of AI-powered SupTech in EMDEs is not merely an IT issue. To harness the three mutually reinforcing advantages of AI, FSAs must undergo fundamental cultural shifts supported by careful change management and broad staff mobilization. As Jeník et al. (2025) note, an innovation-oriented culture is a critical element in enabling FSAs to harness innovation for financial inclusion.

- **Workforce strategy:** FSAs must develop and continuously update a comprehensive workforce strategy that responds to evolving technological and supervisory priorities (Bell et al. 2025). The strategy should foster multidisciplinary teams with both domain expertise and AI proficiency to manage AI risks effectively. It should include policies for training and partnerships, such as with academia, to build a pipeline of AI-literate professionals. External experts can temporarily complement internal capacity, and low-code SupTech tools may be prioritized in certain areas to alleviate skills gaps. FSAs can also consider embedding digital and supervisory transformation competencies in the performance evaluations of staff and leadership.
- **Responsible innovation culture:** Fostering a culture of innovation, while redesigning organizational processes, can maximize the benefits of AI. It is crucial that senior leadership within FSAs lead any transformation process to ensure effective change management. FSAs may need to reshape workflows across the organization, break down departmental silos, improve procurement processes, and foster openness to safe and responsible experimentation through initiatives such as calls for input to identify use cases for AI-powered SupTech, innovation challenges, and innovation labs. Such initiatives should reflect a high level of awareness of the financial inclusion and consumer protection implications of AI outputs, the need for data inclusiveness and fairness, and the value of collaboration with FSPs. While the transformation agenda should be ambitious, the speed of reform should be tempered to mitigate resistance from FSA staff.
- **Foundational supervisory approach:** FSAs in EMDEs should accelerate the shift toward effective risk-based supervision, using AI tools as catalysts—not substitutes—for foundational supervisory capacity, as noted by Dohotaru et al. (2025). A forward-looking, risk-based mindset may require training in supervisory judgment informed by AI outputs. AI-powered SupTech should also be used to automate and modernize supervisory processes following a careful review of supervisory procedures and manuals. For example, AI-powered SupTech is integral to several market-monitoring tools that enable FSAs to gather deep insights into consumers' experiences, risks, and outcomes when using financial services and to strengthen risk-based conduct supervision (Izaguirre et al. 2022). Cross-departmental, multidisciplinary groups can help identify priority use cases; one example is the Applied Technology Laboratory and Experiment Department at the Bank of Spain, which functions as a cross-departmental hub dedicated to developing AI capabilities by leveraging multidisciplinary teamwork.

5. LEVERAGE DOMESTIC AND INTERNATIONAL COLLABORATION

Collaboration and experimentation at domestic and international levels can help FSAs find solutions to challenges in adopting AI-powered SupTech. An example of domestic collaboration includes the joint initiative between the Bank of England and the Financial Conduct Authority to transform data collection (BoE and FCA 2022). The Bank of England is also establishing an Artificial Intelligence Consortium to

bring together public and private stakeholders to advance responsible AI use, similarly to the FREE-AI Committee created by the Reserve Bank of India (Sankar 2025). Additionally, FSAs in EMDEs note the need to collaborate with data protection and cybersecurity authorities (World Bank 2025). Collaboration among authorities, providers, consumer representatives, and market facilitators is key to addressing evolving AI-related consumer risks and building a responsible digital finance ecosystem that places customers at the center (Duflos et al. 2024).

Internationally, the BIS Innovation Hub supports cooperation and peer learning through specialized projects (e.g., Project AISE—Artificial Intelligence Supervisory Enhancer), and the Global Financial Innovation Network (GFIN) offers a knowledge exchange forum for FSAs including in EMDEs.

International cooperation can also help find solutions to the issues raised by the concentration of AI capabilities among a small number of global firms as noted by Garvey et al. (2024). Additional collaboration models, such as shared infrastructure and supervisory colleges, should be explored to help FSAs in EMDEs balance the risks and opportunities of AI-powered SupTech. Shared resources may be particularly attractive for small FSAs and those facing severe budgetary constraints in EMDEs. One example is the Banking Supervision Application, shared by 21 central banks in Africa and Asia, to which AI capabilities will be added (AFI 2024).

Conclusion

Fully leveraging AI-powered SupTech in EMDE FSAs demands profound organizational and cultural transformation to ensure sustained capability improvements in a responsible manner. FSAs need institution-wide strategies to address foundational infrastructure, systems, governance, and capacity issues and to transform regulatory and supervisory processes while leveraging AI on that journey. They need to cultivate an organizational culture that is innovative, attentive to responsible financial inclusion and consumer outcomes, and supported by workforce strategies aligned with SupTech strategies. FSAs must manage the risks associated with AI adoption while weighing them against the risks and limitations of existing methods. Strong data and AI governance are essential to ensure the ethical, fair, and transparent use of granular data and to mitigate the risk of reinforcing bias and exclusion.

Crucially, the promise of AI-powered SupTech in EMDEs cannot be realized without addressing persistent data challenges. A future-proof approach to AI-powered SupTech is anchored in an ambitious but realistic digital transformation that addresses foundational weaknesses such as poor-quality data. Most FSAs in EMDEs will need to substantially improve their data and IT infrastructure to ensure that AI supports sound regulatory and supervisory decisions based on inclusive granular data that adequately cover historically underserved markets; in some cases, this will require reviewing the FSA's regulatory perimeter. Equally critical is the development of a strong legal foundation for the use of granular data, AI tools, and infrastructure such as cloud computing in support of an inclusive and responsible financial ecosystem.

FSAs in EMDEs must deploy AI-powered SupTech in ways that foster responsible financial inclusion through the transmission channels identified in this paper. As AI continues to evolve rapidly in the coming years, failure to adopt it in regulatory and supervisory processes may widen the gap between FSAs and FSPs, progressively reducing supervisory effectiveness. Some FSAs may be able to leapfrog, while others may need to take more gradual steps. In either scenario, the objective should not be merely to use AI-powered SupTech, but to leverage it to foster a more inclusive and responsible financial system that meets the needs of all consumers and promotes financial health.

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ANNEX 1:

Analysis of the Linkages Between AI-powered SupTech and Financial Inclusion

Support Functions & Their Financial Inclusion Linkages	
Ensuring Availability and Usability of Quality Data	Supports advanced regulatory and supervisory analytics, and information dissemination activities, underpinning all statutory goals and increasing public trust in the FSA's ability to fulfil them. Modern data collection, validation and management systems at FSAs can speed up the onboarding of new FSPs (which can support competition) and make IT audits and other examinations more agile and precise, reducing compliance costs .
Improving Customer- and FSP-Facing Services (Including Complaints Handling)	Breaks language, mobility, literacy and other barriers for customer engagement with FSAs, increasing trust , and enables fair complaint resolution, supporting consumer protection . Also reduces barriers and costs for FSPs to obtain data and information produced by FSAs, supporting business strategies and reducing time-to-market for new licensing applicants, ultimately supporting competition .
Financial Inclusion and Health Monitoring and Dissemination	Directly improves financial inclusion monitoring and policymaking, also supporting consumer protection via financial health monitoring.
Monitoring Public Trust	FSA's reputation management is key for trust and credibility, which supports regulatory compliance and financial stability . Accurate assessment of public trust can help FSA better coordinate and implement complex reforms such as open finance and fast payment systems that foster competition .
Enhancing Public Dissemination and Communication	Helps consumers, investors and others obtain information about FSPs, products and services. Also ensures FSAs' consistency of messaging, and broadens the reach of their communications to make information accessible and understandable, contributing to regulatory certainty and trust . These can also reduce information asymmetry, supporting competition .
Talent Management	A diverse and skilled workforce can foster innovation culture and fight unconscious gender and other biases in regulatory and supervisory actions, supporting trust and competition . A more

diverse financial system is more likely to support inclusion (Sotiriou et al. 2024) and be more sustainable (Gambacorta et al. 2022).

Regulatory Functions & Their Financial Inclusion Linkages	
Assessing Regulatory Effectiveness and Impact	Helps FSAs identify the impact of regulation to support all their statutory goals (e.g., financial stability, financial markets integrity, consumer protection, competition, financial inclusion).
Improving Regulatory Framework Management	Increases transparency and regulatory certainty for FSPs, encouraging investors and supporting competition , and reduces compliance costs by ensuring proportionality and relevance of regulatory requirements.
Streamlining the Regulatory Reform Process	Supports trust in FSAs by increasing transparency and accountability; increases regulatory certainty, encourages investors to support greater competition ; and reduces compliance costs via alignment of regulatory interpretation between FSAs and FSPs and across the market.
Machine-Executable Regulation	Increases regulatory certainty which fosters competition and reduces compliance costs such as those attached to regulatory reporting, in the long run.
Supervisory Functions & Their Financial Inclusion Linkages	
Cross-Functional Processes and Procedures (e.g., advanced text analysis, supervisory scheduling, staff allocation, workflow integration, supervisory tool development).	Automation of supervisory processes that involve manual and repetitive tasks (Dohotaru et al. 2025) are all high-value use cases that indirectly support financial inclusion by increasing efficiency of supervision, reducing compliance costs for FSPs and increasing proportionality of supervision, which could help level the playing field and support competition . They also increase accountability and transparency about FSA's supervisory approach, increasing trust .
Enhancing and Streamlining Licensing and Authorizations	Increases transparency, eases market entry for FSPs including challengers, supporting competition and reducing ongoing compliance costs (eg, product approvals).
Prudential Supervision (Banks, nonbanks, payments, insurance, capital markets): Microprudential supervision and macroprudential oversight	Enhanced prudential supervision protects retail depositors, increasing trust and leading to greater engagement with the financial sector. SupTech helps FSAs identify and address risk dynamics and risk build up quicker and more effectively, increasing risk sensitivity for better allocation of supervisory intensity and focus. This leads to more proactive, pre-emptive and accurate supervisory action, improving risk management and mitigation by FSPs, and supporting financial stability and resilience , while reducing compliance costs . More direct links exist in some risk areas. For instance, improving cyber defenses by FSPs reduces the risk of consumer loss due to data

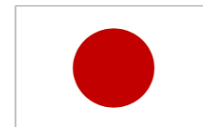
	breaches, increasing trust . Improved credit risk and insurance management could support inclusion and consumer protection .
Payments Oversight: Large value payments and retail payments	Technology-enhanced payments oversight directly supports financial stability and resilience . Retail payments supervision ensures the good functioning and security of low-cost and accessible digital payments, and detection and reduction of frauds, which support financial inclusion via increased acceptance and usage.
Market Conduct Supervision (Financial integrity, consumer protection and competition)	Enhanced conduct supervision increases trust in the financial sector and levels the playing field, supporting competition . AML/CFT supervision reduces the risk of consumer accounts being used to support financial crimes, increasing financial market integrity . Targeted assessment of risk-based implementation of customer due diligence requirements helps fight overcompliance that can impact financial inclusion . Better supervision improves consumer protection , ensuring fair treatment, reducing risks such as misselling, over-indebtedness, discrimination, and financial losses. Competition-focused supervision encourages fair competition , enabling entry of inclusive service providers, and ensuring a level playing field, with indirect financial inclusion impacts.
Monitoring Unregulated Markets	Improved surveillance of unregulated markets contributes to financial stability, competition, inclusion and consumer protection , as it supports evidence-based reviews of the regulatory perimeter, bringing under the regulatory purview FSPs serving large numbers of customers or with important interconnections with regulated FSPs. Improving identification of financial scams such as unauthorized providers has consumer protection and trust benefits by reducing the losses, taking timely measures against perpetrators, and customizing consumer awareness campaigns.

Source: Authors

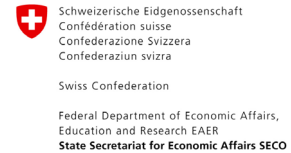
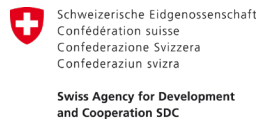
CGAP Members



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CGAP Members (continued)



CGAP Strategic Partners





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